FSK116 Sem Test 2 Draft 2 for 24 May 2022 Final after WEM MEMO

Physics for Engineers (University of Pretoria)

PLEASE PRINT YOUR DETAILS, NEATLY!
PULL- OFF THIS PAGE FROM THE STAPLE AND SEND IT TO THE EDGE OF YOUR BENCH, AS DIRECTED BY THE INVIGILATORS

Surname and Names

Student Number

Faculty of Natural and Agricultural Sciences
Fakulteit Natuur- en Landbouwetenskappe
Lefapha la Disaense tša Tlhago le Temo

Universiteit van Pretoria
University of Pretoria
Yunibesithi ya Pretoria

Department of Physics

FSK116
SEMESTER TEST 2

Examiners: Mr MJ Legodi
Moderator: Prof WE Meyer

Date: 24 May 2022
Time: 13:30 to 16:00

Total 60

FSK116 Semester Test 2

Please read and understand the following general rules.

1. Your handwritten script will be scanned into our Gradescope, after you hand it in to the invigilators.
2. Be seated at the test venue fifteen (15) minutes before the start of the test.
3. All test and examination regulations of the University of Pretoria apply.
4. Answer the questions in your own handwriting, in the spaces provided.
5. Number and label your answers as in the question paper. Incorrectly-numbered and incorrectly-labelled answers may not be marked.
6. Write your answers only on one side of the question paper. Your work will not be marked otherwise. At the end of the question paper, we have made provision for overflows. Clearly indicate which question or subquestion it is you are continuing in the overflow pages.
7. The invigilators will hand out attendance slips for this test, fill them up immediately and send them to the edge of your bench for collection.
8. Submit your Cheat-sheet in the box designated for Cheat-sheets!
9. If any misconduct with respect to this test is suspected, you will receive a mark of zero and will have to attend a formal disciplinary hearing, which may result in your expulsion from the University of Pretoria.
10. This test consists of a total of 16 pages (including this one).
11. The next page contains specific rules that you ought to read and understand!

Department of Physics, FSK116, Semester Test 2

PLEASE WRITE / PRINT YOUR SURNAME AND NAMES, NEATLY!
(They will be read-in by machine!)

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The following specific rules apply to this Semester test

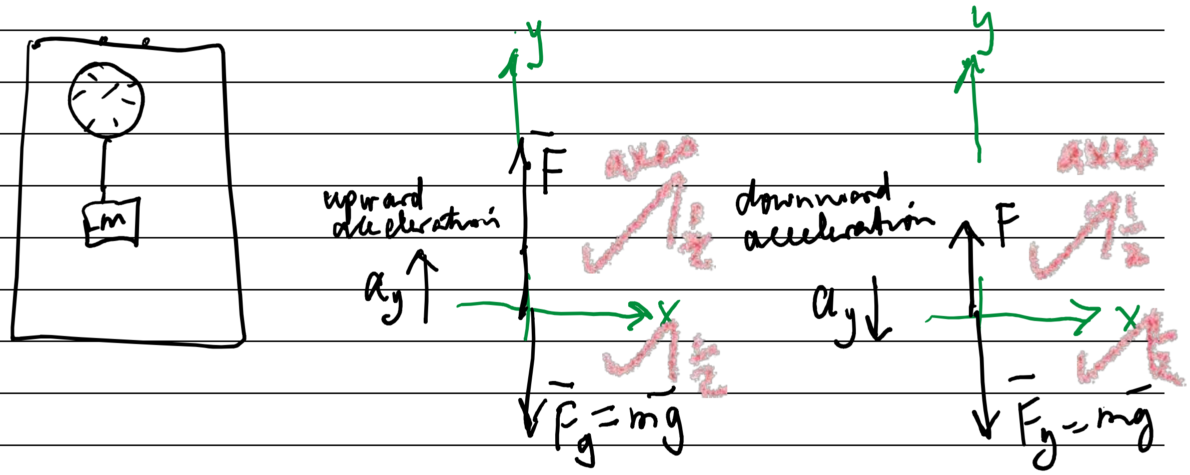
1. All test and examination regulations of the University of Pretoria are applicable.
2. Answer all questions and in the space provided on the question paper. If you need more space, please use the "overflow pages" at the back of the question paper and clearly indicate where the question continues.
3. Pocket calculators may be used, but not programmable calculators.
4. No cell phones are allowed in the test/examination venue.
5. This paper may not be taken out of the test/examination venue. It must be handed in to the chief invigilator, before you leave the venue.
6. Work written in pencil will not be marked. Write in black pen.
7. Please provide labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct significant figures, etc. Failure to do so may result in marks being deducted.
8. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not mean that you will receive full marks.
9. Units must be used when numbers are substituted else marks will be deducted.
10. Extra points will be awarded for high quality or elegant answers.

Constants, conversion factors and some formulae that may be useful

$g = 9.80\ \text{m/s}^2$ $\qquad m_{\text{Earth}} = 5.972\text{x}10^{24}\ \text{kg}$ $\qquad R_{\text{Earth}} = 6.34\text{x}10^{6}\ \text{m}$

Quaestion 1 **[7]**

1.1 In the Natural Sciences 1 building's lift, a visiting Grade 12 learner observes that the reading of the weight of a mass *m* attached to the end of a spring balance (that is in turn attached to the lift ceiling) seems to vary depending on whether the lift is accelerating upwards or downwards. Fortunately for the learner and society at large, you are in the same lift. Develop a calculation that will help the learner understand what is going on in each case. Take the force that the spring balance applies to *m* as **F**. **NOTE**: include free body diagrams and all the laws of Physics that you will utilize in your argument. (7)



Newton's II law applied to both upward and downward motions.

UPWARDS

$\sum F_y = ma_y$

$F - mg = ma_y$

$\therefore F = mg + ma_y$

$= mg\left(1 + \frac{a_y}{g}\right)$

$\sum F_x = ma_x = 0$

There is no motion in x-axis

7

DOWNWARDS

$\sum F_y = ma_y$ and $\sum F_x = ma_x = 0$

$F - mg = -ma_y$

as before, there is no acceleration in the x-axis.

$\therefore F = mg - ma_y$

$= mg\left(1 - \frac{a_y}{g}\right)$

$\therefore$ The apparent weight of m is dependent on g and the acceleration of the lift.

Question 2 **(11)**

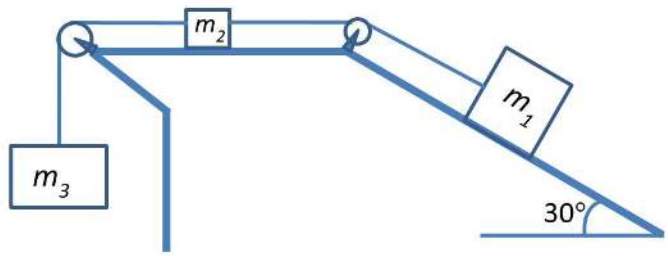
2.1 Consider the diagram below. Three masses are connected to each other by a rope with negligible mass, and the rope passes over two ideal pulleys. The surface on which m_2 moves is smooth, while the coefficient of static friction between m_1 and the inclined surface is 0.300.

If m_1 is 10.0 kg, and m_2 is 1.00 kg,

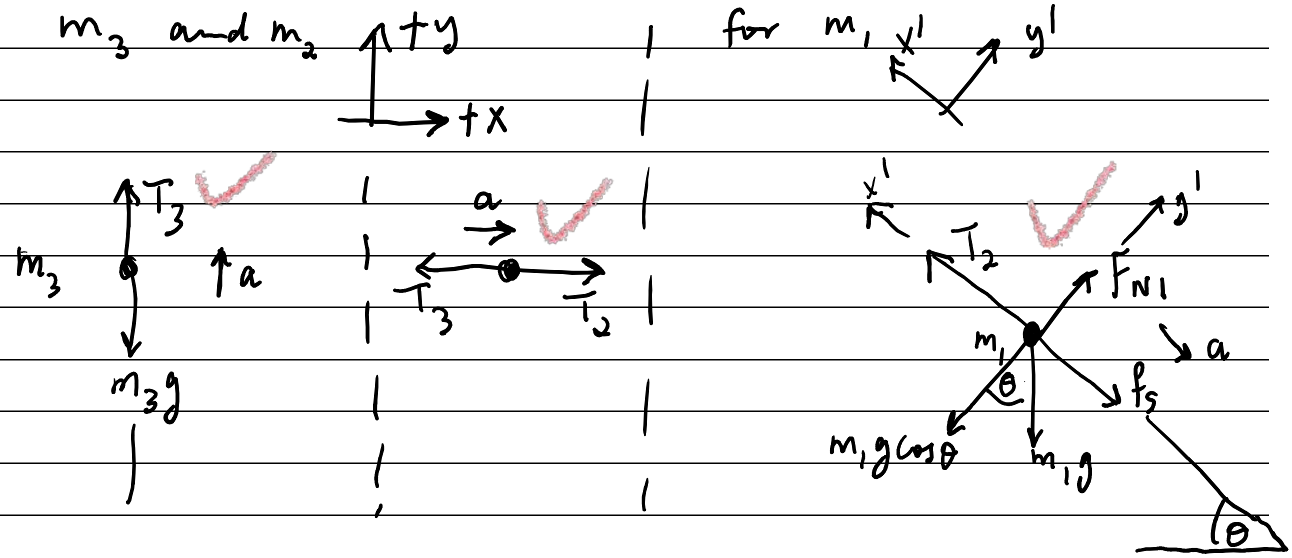
(a) What is the minimum mass of m_3 that will ensure that m_1 starts to slide up the slope when released from rest? (7)

(b) Using the mass m_3 determined in (a) determine the tension in the ropes to the left and right of m_2 when m_1 moves up the incline. (4)

REMEMBER: *State your assumptions and provide reasons for your actions. And, show properly labeled free-body diagrams for all relevant objects.*



FREE-BODY DIAGRAMS



From the free-body diagrams, we have the following:

$m_3 g - T_3 = m_3 a_3 = m_3 a$; $T_3 - T_2 = m_2 a_2 = m_2 a$; $T_2 - m_1 g \sin\theta - f_s = m_1 a_1$
$= m_1 a$

$a_1 = a_2 = a_3 = a$ since we employ an ideal rope of negligible mass.

In fact, $a = 0$ since we are analysing impending motion:

for m_2 and m_3;

$T_3 - T_2 = m_2 a$ --- (A)

$m_3 g - T_3 = m_3 a$ --- (B)

$\Leftrightarrow m_3 g - T_2 = (m_2 + m_3) a$ after adding --- (C)

And $T_2 - m_1 g \sin\theta - f_s = m_1 a$ --- (D)

$\Rightarrow m_3 g - m_1 g \sin\theta - f_s = (m_1 + m_2 + m_3) a$ after adding (C) and (D)

This simplifies to; $m_3 g - m_1 g \sin\theta - f_s = 0$ since we are analysing impending motion when $a = 0$

7

$\Rightarrow m_3 g = m_1 g \sin\theta + \mu_s F_{N1}$

$= m_1 g \sin\theta + \mu_s m_1 g \cos\theta$ after subbing for F_{N1}

$\Rightarrow m_3 = m_1 (\sin\theta + \mu_s \cos\theta)$

$= (10.0\ kg)(\sin 30° + 0.300 \cos 30°)$

$= 7.6\ kg$

minimum mass to let m_1 slide up the incline.

From $m_3 g - T_3 = m_3 a = 0$

$\Rightarrow T_3 = m_3 g$

$= (7.6\ kg)(9.80\ m/s^2)$

$= 75 N$

4

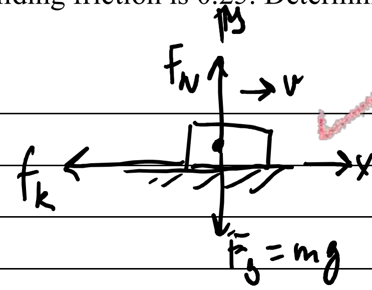
And from $T_3 - T_2 = m_2 a = 0$

$\therefore T_2 = T_3 = 75 N$

Question 3 [15]

3.1 A person pushes a box across a horizontal surface at a constant speed of 0.5 meter per second. The box has a mass of 40 kilograms, and the coefficient of sliding friction is 0.25. Determine the power supplied to the box by the person. (2)

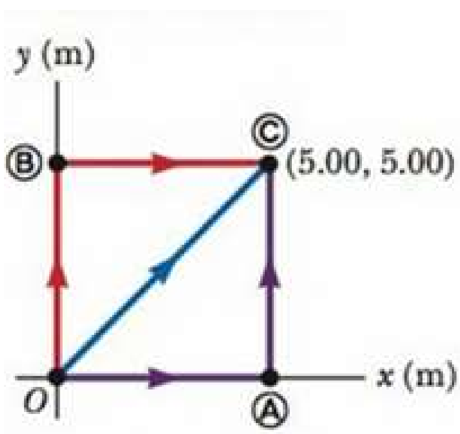
$$p = \bar{F} \cdot \bar{v} = F' v$$
$$= (\mu_k F_N)\, v$$
$$= \mu_k m g v$$
$$= (0.25)(40\,kg)(9.80\,m/s^2)(0.5\,m/s)$$
$$= 49\ kg\,m^2/s^3 \text{ or } W$$



F' is force applied to the block!

2

3.2 A 4.00 kg particle moves from the origin to position C, having coordinates x = 5.00 m and y = 5.00 m, as shown in the figure below. One force on the particle is the gravitational force acting in the negative y direction. Using $W = F\Delta r \cos\theta$, calculate the work done by the gravitational force on the particle as it goes from O to C along…



(a) the path OAC, (2)

From $W = F\Delta r \cos\theta$ with θ the angle between the force F and displacement Δr,

$$F_g = mg = (4.00\,kg)(9.80\,m/s^2) = 39.2\,N$$

Work along OAC = Work along OA + W along AC
$$= F_g (OA) \cos 90.0° + F_g (AC) \cos 180.0°$$
$$= 39.2\,N\,(5.00\,m)(0) + (39.2\,N)(5.00\,m)(-1)$$
$$= -196\,J$$

2

(b) path OBC, and (2)

Work along OBC = Work along OB + Work along BC

$= (39.2N)(5.00m)(\cos 180.0°) +$

$(39.2N)(5.00m)(\cos 90.0°)$

$= -196\ J$

2

(c) the path OC (2)

Work along OC = $F_g\,(OC) \cos 135°$

$= (39.2N)(5.00m)(\sqrt{2})\left(-\frac{1}{\sqrt{2}}\right)$

$= -196\ J$

2

(d) Your results should all be identical. Why? (2)

The results are all the same since gravitational force is conservative ie. its line/path integral $\int \bar{F}_g \cdot d\bar{l}$, which is the basis for all the calculations above ie $\int \bar{F} \cdot d\bar{r} = F \Delta r \cos\theta$ is path independent.

2

(e) Suppose the force $\vec{\mathbf{F}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ N acts on a particle that moves from O to C as shown in the figure. Use $W = \int_i^f \vec{\boldsymbol{F}} \cdot d\vec{\boldsymbol{r}}$ to calculate the work done by $\vec{\mathbf{F}}$ on the particle. (5)

$$W = \int_i^f \bar{F} \cdot d\bar{l} = \int_i^f (3\hat{i} + 4\hat{j}) \cdot (dx\,\hat{i} + dy\,\hat{j})$$

$$= \int_0^{5.00m} 3\,dx + \int_0^{5.00m} 4\,dy$$

$$= (3\,N)x\Big|_0^{5.00m} + (4\,N)y\Big|_0^{5.00m}$$

$$= (15 + 20)\ J$$

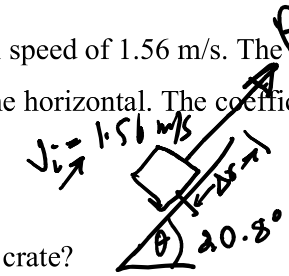
$$= 35\ J$$

All paths result in the same answer for Work done

5

Question 4 [10]

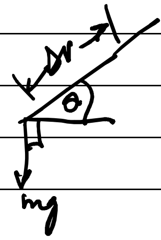
4.1 A crate of mass 9.2 kg is pulled up a rough incline with an initial speed of 1.56 m/s. The pulling force is 92 N parallel to the incline, which makes an angle of 20.8° with the horizontal. The coefficient of kinetic friction is 0.400, and the crate is pulled 5.02 m.



$m = 9.2$ kg
$\mu_k = 0.400$
$\Delta r = 5.02$ m

(a) How much work is done by the gravitational force on the crate? (2)

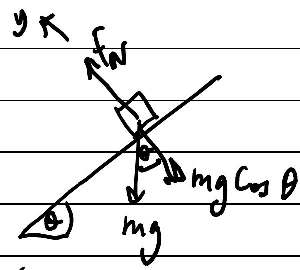
$$W_g = \vec{F}_g \cdot \Delta\vec{r}$$
$$= (mg)(\Delta r)\cos(90^\circ+\theta)$$
$$= -mg\,\Delta r \sin\theta$$
$$= -(9.2\,kg)(9.80\,m/s^2)(5.02\,m)(\sin 20.8^\circ)$$
$$= -160.72\,J$$
$$= -161\,J$$



2

(b) Determine the increase in internal energy of the crate–incline system owing to friction. (2)

$$\Delta E_{int} = f_k d$$
$$= \mu_k F_N d$$
$$= \mu_k (mg\cos\theta) d$$
$$= (0.400)(9.2\,kg)(9.80\,m/s^2)(\cos 20.8^\circ)(5.02\,m)$$
$$= 169.24\,J$$
$$= 169\,J$$



$\Sigma F_y = ma = 0$
$\Rightarrow F_N = mg\cos\theta$

2

(c) How much work is done by the 92-N force on the crate? (2)

We will identify the pull force as P.

$\therefore W_P = |P||\Delta r|\cos\phi$, ϕ is angle between P and the displacement of the crate, d.

$= (92N)(5.02m)(1)$

$= 461.84 J$

$= 4.6 \times 10^2 J$

2

(d) What is the change in kinetic energy of the crate? (2)

$W_{other} = \Delta K + \Delta U_g + \Delta E_{int}$

In our case, W_{other} encompasses work done by force P and work done by F_N. Gravitational force and frictional force have already been accounted for on the RHS of the given equation.

$\therefore \Delta K = W_{other} - \Delta U_g - \Delta E_{int} = W_P - (mgy_f - mgy_i) - \Delta E_{int}$

$= 461.84J - 160.7225J - 169.242037J$

$= 131.875J$

$= 132J$

2

Aside: $\Delta U_g = mgy_f - mgy_i = mgd\sin\theta - 0$

$= 160.7225J$

(e) What is the speed of the crate after being pulled 5.02 m? (2)

$\Delta K = K_f - K_i$

$\Rightarrow K_f = \Delta K + K_i$

$\frac{1}{2}mv_f^2 = \Delta K + \frac{1}{2}mv_i^2$

$\therefore v_f = \sqrt{\frac{2}{m}\left(\Delta K + \frac{1}{2}mv_i^2\right)}$

2

$= \sqrt{\left(\frac{2}{9.2kg}\right)\left[131.875J + \frac{1}{2}(9.2kg)(1.56 m/s)^2\right]}$

$= 5.5768 m/s$

$= 5.6 m/s$

Question 5 **[9]**

5.1 In many calculations that involve a small object and Earth, we often assume that the Earth does not move at all during the interaction. We are convinced that, even if Earth moves, its speed and displacement are extremely small, providing justification for the assumption. If we perform an experiment involving dropping a 1.0 kg ball from 1.00 m above the Earth's surface:

(a) Estimate the acceleration of the Earth during the time interval between the release of the ball and impact of the ball with the Earth surface. (3)

(b) Estimate the displacement of the Earth during this time interval. (3)

(c) Determine an estimate of how fast is the Earth moving when it encounters the ball. (3)

(a) Suppose we drop the ball down to Earth as suggested. From Newton III, we have $|F_E| = |F_B|$

$$\Rightarrow \quad m_E a_E = m_B a_B$$

$$\therefore \quad a_E = \frac{m_B}{m_E} a_B$$

$$\approx \left(\frac{1.0\,kg}{10^{25}\,kg}\right)(9.80\,m/s^2)$$

$$\approx 10^{-24}\,m/s^2$$

3

(b) $y_{f,E} - y_{i,E} = v_{i,E}\Delta t + \frac{1}{2} a_E (\Delta t)^2$ for Earth, and

$y_{f,B} - y_{i,B} = v_{i,B}\Delta t + \frac{1}{2} a_B (\Delta t)^2$ for the ball.

The time intervals are equal, therefore:

$$\frac{2(y_{f,B} - y_{i,B})}{a_B} = \frac{2(y_{f,E} - y_{i,E})}{a_E}$$

3

$$\Rightarrow (y_{f,E} - y_{i,E}) \approx \left(\frac{a_E}{a_B}\right)(y_{f,B} - y_{i,B})$$

$$= \frac{10^{-24}\,m/s^2}{10\,m/s^2}(1\,m) \approx 10^{-24}\,m$$

(c) We can estimate the time it takes for the 1 kg ball to travel under gravity to reach Earth about 1 m below release. We determine the time interval involved, Δt:

$$(y_{f,B} - y_{i,B}) = v_{i,B}(\Delta t) + \frac{1}{2}(-9.80\,m/s^2)\Delta t^2$$

$$\Rightarrow \Delta t^2 \approx 2(y_{f,B} - y_{i,B})/(-9.80\,m/s^2) \approx 2(-1\,m)/(-9.80\,m/s^2)$$

$$\approx 453 \times 10^{-3}\,s$$

3

$$\therefore v_{f,E} = v_{i,E} + a_E(\Delta t)$$

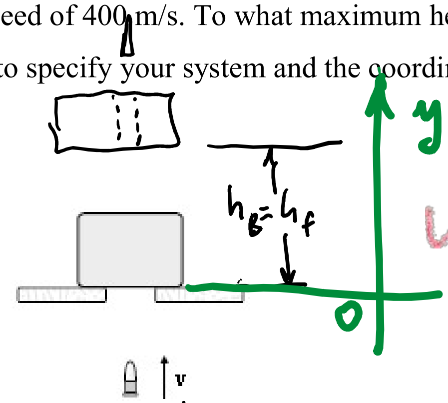
$$\approx (10^{-24}\,m/s^2)(453 \times 10^{-3}\,s) \text{ for } v_{i,E} = 0$$

$$\approx 4.53 \times 10^{-25}\,m/s$$

$$\sim 10^{-24}\,m/s$$

Question 6 **[8]**

A 10.00 g bullet moving 1 000 m/s strikes and passes through a 2.00 kg block initially at rest, as shown. The bullet emerges from the block with a speed of 400 m/s. To what maximum height will the block rise above its initial position? ***NOTE***: you need to specify your system and the coordinate axes involved in the calculation. (8)



$m_b = 10.00\,g$ $\quad$ $m_B = 2.000\,kg$

$v_{b,i} = 1000\,m/s$ $\quad$ $v_{B,i} = 0\,m/s$

$v_{b,f} = 400\,m/s$ $\quad$ $h_B = h_f = ?$

8

Choose system to be bullet + block

From momentum conservation, we have:

$$m_b v_{b,i} + m_B v_{B,i} = m_b v_{b,f} + m_B v_{B,f}$$

$$\therefore v_{B,f} = \frac{m_b v_{b,i} - m_b v_{b,f}}{m_B} \quad \text{since } v_{B,i} = 0$$

$$= \frac{(10.00 \times 10^{-3}\,kg)(1000\,m/s) - (10.000 \times 10^{-3}\,kg)(400\,m/s)}{2.00\,kg}$$

$$= 3.00\,m/s \text{ upwards}$$

The kinetic energy of the block will be converted to gravitational potential energy as the block rises

We could apply the Work-Kinetic Energy theorem to determine the height that the block rises to.

$W_{ext} = \Delta K$ for the block as a system now

$\therefore \vec{F} \cdot \Delta \vec{y} = K_f - K_i$

$|m_B g||y_f - y_i| \cos\theta = 0 - K_i = -\frac{1}{2} m_B v_{B,f}^2$

We have since determined $v_{B,f}$ which defines, in this system consisting of the block only, the 'initial' kinetic energy of the system.

$$\therefore -m_B g h_f = -\frac{1}{2} m_B v_{B,f}^2$$

$$\therefore h_f = \frac{\frac{1}{2} m_B v_{B,f}^2}{m_B g}$$

$$= \frac{(3.00 \text{ m/s})^2}{2(9.81 \text{ m/s}^2)}$$

$$= 45.9 \text{ cm}$$

END

Good Luck!

OVERFLOW PAGES FOLLOW IN THE NEXT TWO PAGES!